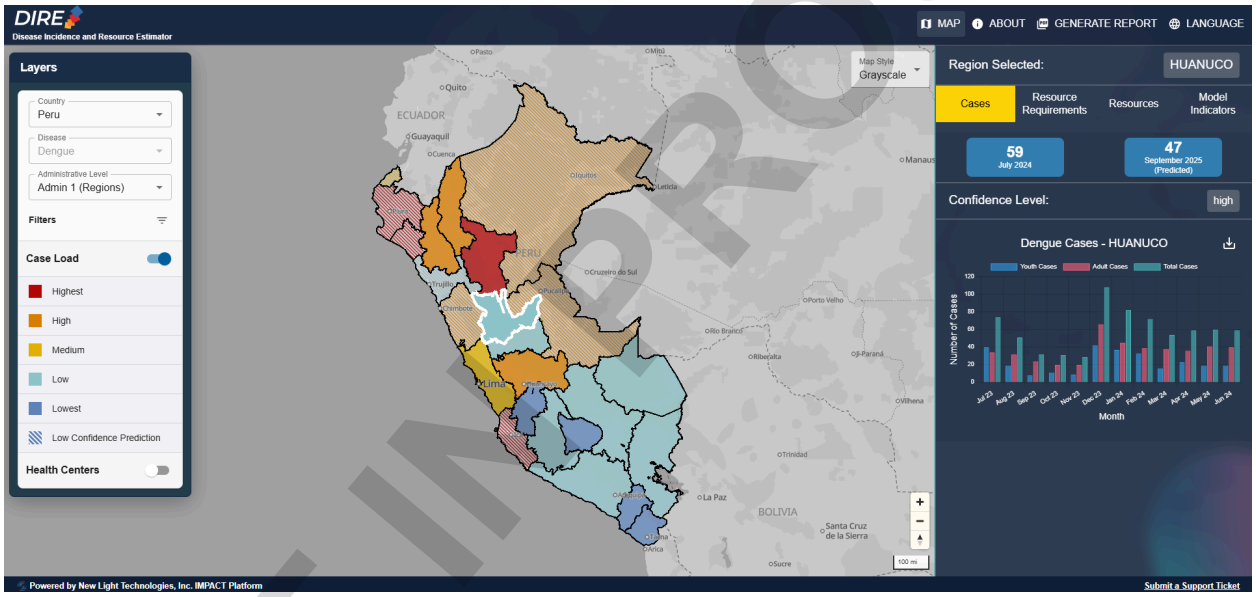
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NLT New Light Technologies

White Paper / Methodological Note

DIRE

The Disease Incidence and Resource Estimator (DIRE)



https://dire.impact.nltglobal.com/

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1. Background

Dengue is a viral infection transmitted to humans through the bite of infected mosquitoes, primarily *Aedes aegypti* (and to some extent by *Aedes albopictus*). These species have adapted to human-inhabited areas, utilizing natural and artificial water sources for oviposition and larval development in urban and peri-urban settings [1]. In urban areas, Dengue is vectored by these mosquitoes that breed in and around human habitats, placing billions of humans at risk of infection with one or more of four Dengue viruses types [2].

Over the past few decades there has been a rise in the number of people infected by Dengue. According to the World Health Organization (WHO) the number of Dengue cases increased from 505,430 in 2000 to 14.6 million in 2024 [3]. The CDC estimates that every year up to 400 million people are infected by a Dengue virus [4].

1.1. *The Spatial Distribution of Dengue*

Dengue is now endemic in over 100 countries, with the highest prevalence in the tropical and subtropical regions of Southeast Asia, Africa, the Western Pacific, and the Americas [5]. In North and South America in particular, the number of dengue cases has risen over the last four decades [6]. However, *Aedes* mosquito vectors are gradually broadening the virus's geographical distribution [7], reaching areas that were not previously affected, including in non-tropical regions [5]. The expanding range of the mosquito vector, affected in part by climate change, increases risk in new areas such as Spain, Portugal, and the southern USA. Emerging evidence points to silent epidemics in Africa [8]. Rapid urbanization, and socio-economic conditions in these affected areas create an ideal environment for the *Aedes* mosquito [5].

In 2024, Brazil was responsible for more than 50% of suspected cases and deaths from Dengue compared to the other Latin American countries [9]. That year, the country experienced a fourfold increase in Dengue cases compared to 2023, accompanied by higher mortality [5]. The increased incidence of Dengue in central Brazil, particularly between 2014 and 2020, can be attributed to the previous circulation of the virus, urbanization, and the occurrence of temperature anomalies for a prolonged period of time [10]. Dengue is also currently the most significant vector-borne disease in Peru, with incidence increasing markedly over the past few years. Overall, Dengue is

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expanding across three primary geographic areas in Peru: the Amazon rainforest, the northern coast, and the central and southern coast [11]. The rise in dengue cases in the country may be linked to a complex combination of factors such as trade, globalization, population movement, demographic shifts, and, most significantly, climate change [12].

1.2. *Environmental Factors Contributing to the Expansion of Dengue*

Various environmental factors have been shown to increase the risk of Dengue infection. Temperature, for example, plays a vital role in the development and survival of *Aedes* mosquitoes [13]. Studies suggest that the risk of dengue infection increases by 13% for each 1°C increase in temperature [14]. Factors such as relative humidity and precipitation also influence the occurrence of Dengue fever [15], [16], [17]. A relative high humidity may prevent mosquito habitats and adult mosquito from drying up [18], while rainfall can increase water storage and vegetation, aiding mosquito oviposition and larval growth [19]. Thus, extreme rainfall can contribute to the spread of mosquito-borne diseases by generating breeding habitats, damaging infrastructure, and disrupting vector control efforts [20]. These factors make wet bioclimatic zones more at risk to Dengue infections than dry zones [17]. Moreover, in the last few decades, many developing countries have been experiencing an increase in natural disasters such as floods, tsunamis, and storms, which exacerbate the formation of standing water in large urban areas, facilitating the spread of dengue and other mosquito-borne diseases [5].

Urban areas and urbanization are also linked to Dengue outbreaks, with urbanization positively associated with Dengue incidents and spread [21]. In addition, inadequate sanitation, poor waste management, and lack of access to clean water can increase the risk of Dengue transmission [22]. Lack of access to tap water often results in greater dependence on water storage containers, which can create favorable conditions for mosquito breeding [23].

1.3. *Dengue Prevention and Mitigation*

Overall, good prevention practices in the community are key for minimizing the risk of Dengue fever outbreaks. In this respect, studies call for a multifaceted approach that considers climatic, environmental, and behavioral factors, together with tailored strategies for different geographical areas [12]. Education, knowledge about Dengue

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prevention, and mass media education, are all essential for the prevention of Dengue fever [24]. Other control strategies include vaccines, antivirals therapies, improved environmental sanitation, as well as measures taken to manage or prevent the deposition of eggs and larvae in water accumulations, tires, plant pots, and uncovered water tanks [25],[26]. Other approaches include elimination of open water sources, spraying insecticides or biological control agents on the affected areas, wearing clothes that completely covers the skin, sleeping with mosquito netting and more [27].

The World Health Organization (WHO) outlines five key components for the effective implementation of an Integrated Vector Management (IVM) program: (1) Advocacy, social mobilization, and regulatory measures to protect public health and empower communities; (2) Collaboration across the health sector and with other sectors to optimize resources, strengthen planning, monitoring, and decision-making; (3) Integration of chemical and non-chemical vector control methods, alongside other disease control measures; (4) Evidence-based decision-making, supported by operational research, entomological and epidemiological surveillance, and evaluation; (5) Human resource development, including training and career structures at both national and local levels, to promote capacity building and ensure effective IVM program management [28].

In addition, timely predictions of Dengue outbreaks through early warning systems (EWSs) are essential for prevention, response and mitigation operations and for improving cost-effective interventions. They can support decision makers in mitigating or preventing disease outbreaks. EWSs allow countries to identify consistently through alarm signals an increased outbreak risk in time and space that could trigger an early response [29].

1.4. *Predicting Dengue Fever Outbreaks by Means of Machine Learning*

Prediction of Dengue outbreaks is complex. As discussed in Sebastianelli et al. (2024), many of the existing dengue modelling applications have been developed at either the local or national level, primarily due to different definitions, procedures and practices applied when collecting dengue data in different countries, leading to inconsistencies that prevent from building generalized models capable of predicting dengue incidence across national boundaries [30].

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Prediction of Dengue can be achieved by means of statistical models. Previous studies propose a wide range of statistical techniques for analysis and forecasting of Dengue. These include, for example, time-series analysis with Autoregressive Integrated Moving Average (ARIMA) and Seasonal ARIMA (SARIMA) models [31], Integrated Nested Laplace Approximations (INLA) [32], Exponential Smoothing models [33], General Linear models [34], Poisson and Negative Binomial Regression [35], among others. Time-series regression models have particularly dominated the field of future disease occurrences prediction. These models assume that future values (e.g. predicted Dengue cases) could be predicted based upon past observations and are based upon retrospective determination of statistically significant correlation coefficients [36].

Machine Learning (ML) is also increasingly used to predict Dengue. According to Batista et al. (2021) [37], ML techniques demonstrate a lower error rate compared to the conventional statistics-based models in predicting Dengue. Some of the ML models used in the literature include Random Forests [38], [39], Support Vector Machines (SVM) [40], [41], Gradient Boosting (XXXX), Neural Networks [42], [43] and various ensemble models [30], [44]. In a literature review conducted by [45] across 64 studies that predict Dengue, Random Forests, Neural Networks and ensemble models were the most notable techniques.

1.4.1. *Ensemble Machine Learning*

Studies utilizing ensemble methods use a variety of models, ensembled together. Various ensemble approaches have been proposed in the literature to predict Dengue outbreaks. Such approaches can overcome the limitations associated with classical techniques [46] and address scenarios where single models work well in certain situations and not in others [36]. Loshini et al. (2015) [47] have shown, for example, in the case of Selangor, that ensemble models are able to predict better than single models when accounting for Dengue cases in a district together with its neighboring districts.

Several types of ensemble machine learning approaches have been proposed in the literature to predict Dengue outbreaks. These include Bagging (such as Random Forests), Boosting (e.g., Gradient Boosting, CatBoost), Stacking and Hybrid Ensemble Models.

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Bagging. Like in the case of Random Forests, bagging is used to reduce variance in noisy data. Multiple models are trained independently on a random subset of data (with replacement). The predictions of these models are then aggregated to predict the output of the model. When predicting a numerical outcome, the aggregation averages over the versions and when predicting a class, it does a plurality vote [48].

Boosting. Unlike in the case of bagging, where the models are trained independently and in parallel, in boosting, the models are trained sequentially. Each model learns from the errors of the previous model. The approach`s assumption is that a combination of simple classifiers can perform better than any of the simple classifiers alone. CatBoost, for example, is based on the concept of gradient boosting. Each time a decision tree (a "weak learner") is constructed, prediction errors are calculated and a new tree is created to correct these errors. Multiple "weak learners" (typically decision trees) are combined, sequentially, to create a "strong learner".

Stacking. In Stacking, multiple machine models are combined into one (meta) model to predict the output. This allows leveraging the strength of each model to achieve a more accurate prediction. Numerous base models are trained on the same training dataset and the predictions of these models are used (as inputs) to make a final prediction based on a meta model (which can be, for example, linear regression, logistic regression, random forest etc.). The meta-model is then trained using the predictions given by the base models on a validation set.

Hybrid Ensemble Models. In this type of ensemble, different types of modeling approaches, or heterogenous components, are combined into a single ensemble framework. Such models combine multiple Machine Learning and/or soft computing methods for higher performance and optimum results. They contain multiple different single methods that form a method with higher flexibility and with a high capability compared with single methods [49]. Unlike Stacking ensemble, which generally combines homogeneous models, hybrid models combine different machine learning paradigms (e.g., Machine Learning and Deep Learning; supervised and unsupervised learning etc.).

2. Methodology

2.1. Machine Learning Approach

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In this study, we rely on an ensemble machine learning approach developed by [30] to forecast Dengue outbreaks in two Latin America countries. This model was originally developed to forecast Dengue incidents rate in Brazil and Peru with a focus on the population under 19 years old. The model, which addresses data scarcity and operational scalability challenges in these countries, integrates spatial and temporal information, providing one-month-ahead Dengue Incident Rate (DIR) estimates for each State (per administrative unit). The ensemble comprises three machine learning models, namely CatBoost, Support Vector Machine (SVM) and Long Short-Term Memory (LSTM). The outputs of each ML model are imputed into a Random Forest model.

CatBoost (Categorical Boosting) [50] is a type of Gradient Boosted Decision Trees, which makes use of binary decision trees as base predictors. It is especially useful when handling categorical features and complex and high-dimensional datasets. With CatBoost, categorical features are converted into numerical values automatically without the need for pre-processing or manual feature encoding. In CatBoost, decision trees are built sequentially. For each decision tree, the model evaluates the extent of errors in its predictions; another tree is then built to correct the errors made by the previous one. This process is repeated until a predefined number of iterations are met.

SVM is a classification and regression prediction tool designed to maximize predictive accuracy while avoiding overfit [40]. Originally developed to solve classification problems, the SVM has been extended to solve regression problems [51]. The SVM model consists of two different patterns, and its goal is to separate them. The model consists of three different lines: the marginal line and the lines on either side of the line of margin. These three lines together construct the hyper plane that separates the given patterns. The pattern that lies on the edges of the hyper plane is called support vectors and the perpendicular distance between the line of margin and the edges of the hyper plane is known as margin. One of the objectives of SVM is to maximize this margin for a more accurate classification [52]. Qaiser et al., 2024 [53], for example, have shown that SVM is a superior model in the prediction of Dengue Fever due to its ability to handle nonlinear relationships in high-dimensional spaces effectively.

When solving a regression classification problem, the model is referred to as a Support Vector Regression (SVR). The SVR aims to minimize the coefficients of a defined loss

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function, penalizing high and low mis-estimates and providing flexibility on how much error is acceptable in the model [30]. The SVR identifies the best-fit line (hyperplane) that approximates the data while ensuring that the prediction error is within a certain tolerance.

LSTM [54] is a type of Recurrent Neural Network (RNN) which can be used to model sequential data (where the order of the data points is important) and deal with long-range dependencies. It addresses the vanishing gradient problem associated with RNN, where the gradients used to update the network`s weights become extremely small over time. Some of the advantages of LSTM include its ability to handle noise, distributed representations and continuous values. The unique architecture of LSTM allows them to retain and utilize information over extended periods, making them suitable for time-series prediction and modeling dynamic systems [55]. The LSTM makes use of gating mechanisms to mitigate exploding and vanishing gradients when learning long-term dependencies, making them suitable for sequence modeling tasks [56].

2.2. Dengue Predictors

As described above, factors associated with Dengue outbreaks include climate (like temperature and precipitation), demographic and vector factors [45]. Buczak et al. (2012) [57] assess several factors that can potentially predict Dengue, including previous dengue incidence, meteorological/climatic data (rainfall, day temperature, night temperature), leaf area indices (NDVI, EVI) that may serve as indicators of precipitation, other indicators related to climate effects such as El Nino (the Southern Oscillation Index (SOI) and various sea surface temperature anomalies (SSTA)), and socio-economic indicators (e.g., political stability, sanitation, water, and electricity). Benedum et al. (2020) [58] investigate multiple weather data predictors of Dengue, including temperature, rainfall and humidity. Similarly, [59] estimate the impacts of different short and long-term lagged climatic predictors on Dengue cases, including NDVI, precipitation, temperature and humidity.

In this study, we predict Dengue incidents by utilizing a range of environmental, climatological, geo-physical, socio-economic, epidemiological and population attributes. These include air temperature, dew point, surface pressure, precipitation, wind

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components, NDVI, elevation and forest cover loss, along with socio-economic variables. We utilize six groups of socio-economic variables, namely, House Type, Electricity, Literacy Rate, Population, Water Source and Sewage System (see Table 1).

Table 1: the variables used to predict Dengue outbreaks

Epidemiological and population factors
Dengue cases (total), Dengue cases age-group (0–19 years), Total Population, Population age-group (0–19 years)
Climatic
2m Air Temperature (K) – min, mean, max, 2m Dew Point Air Temperature (K); Surface Pressure (Pa); Total Precipitation (m); 10m u/v Wind components (m/s)
Geophysical
Normalized Difference Vegetation Index (NDVI); Elevation (m) – min, mean, max; Forest Loss / Cover (%)
Socio-economic
House Type, Electricity, Literacy Rate, Population, Water Source and Sewage System

2.3. *Data*

Data used to predict Dengue cases is extracted from countries` national censuses (e.g., Brazil 2010 & 2022, Peru 2007 & 2017) and external repositories for climate and geographic data. Dengue case data is sourced from national health systems (Brazil's DATASUS API - available in https://github.com/rfsaldanha/microdatasus), Peru's XXXX).

2.3.1. *Socio-economic data*

We retrieve socio-economic data from the most recent census surveys available for each country. The socio-economic variables used for Brazil and Peru are presented in table 2.

Table 2: Socio-economic variables used for prediction of Dengue outbreaks

Variable name	**definition**
Peru	
watersource_publicnetworkinhouse	Water source: public network in house
watersource_publicnetworkouthouse	Water source: public network out of house

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watersource_pylon	Water source: Pylon
watersource_truck	Water source: Truck
watersource_well	Water source: Well
watersource_river	Water source: River
watersource_other	Water source: Other
watersource_neighbor	Water source: neighbor
withelectricity	With electricity
withoutelectricity	Without electricity
sewage_river	Sewage: River
sewage_publicnetworkouthouse	Sewage: Public network out of house
sewage_publicnetworkinhouse	Sewage: Public network in house
sewage_septictank	Sewage: Septic tank
sewage_other	Sewage: Other
sewage_rudimentaryseptic	Sewage: Rudimentary septic system
housetype_apartment	House type: Apartment
housetype_collective	House type: Collective
housetype_fifth	House type: Fifth
housetype_house	House type: House
housetype_hut	House type: Hut
housetype_improvized	House type: Improvized
housetype_nonhuman	House type: Non human
housetype_other	House type: Other
housetype_tenement	House type: Tenement
Brazil	
watersource_generalnetwork	Water source: General network
watersource_watertruck	Water source: Water truck
watersource_river	Water source: River
watersource_well	Water source: Well
watersource_other	Water source: Other
withelectricity	With electricity
withoutelectricity	Without electricity
sewage_water	Sewage: Water
sewage_generalnetwork	Sewage: General network
sewage_ditch	Sewage: Ditch
sewage_none	Sewage: None
sewage_septic	Sewage: Septic

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sewage_other	Sewage: Other
sewage_rudimentarypit	Sewage: Rudimentary pit

Socio-economic data for Brazil is based on the countries` 2010 and 2022 censuses found at the following sites:

- 2010: https://sidra.ibge.gov.br/pesquisa/censo-demografico/demografico-2010/universo-caracteristicas-da-populacao-e-dos-domicilios
- 2022: https://sidra.ibge.gov.br/pesquisa/censo-demografico/demografico-2022/universo-caracteristicas-dos-domicilios

Socio-economic data for Peru is retrieved from Peru's 2007 and 2017 censuses found on these sites:

- 2007: http://censos1.inei.gob.pe/Censos2007/redatam/
- 2017: https://censos2017.inei.gob.pe/redatam/

2.3.2. *Climatological, environmental and geographic data*

Climatological, environmental and geographic data are retrieved from external sources, including ERA5-Land hourly dataset, Terra Moderate Resolution Imaging Spectroradiometer (MODIS) Vegetation Indices Monthly (MOD13A3) Version 6.1, Terra and Aqua combined Moderate Resolution Imaging Spectroradiometer (MODIS) Land Cover Type (MCD12Q1) Version 6.1 and NASADEM (through USGS OPeNDAP) (the indicators and their sources are provided in Table 3).

Table 3: Climatological, environmental and geographic attributes used to predict Dengue outbreaks

Indicator	Source
2m temperature	ERA5-Land hourly dataset
2m dewpoint temperature	
10m u component of wind	
10m v component of wind	
Surface pressure	
Total precipitation	

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Normalized Difference Vegetation Index (NDVI)	Terra Moderate Resolution Imaging Spectroradiometer (MODIS) Vegetation Indices Monthly (MOD13A3) Version 6.1
Land Use/ Land Cover (Barren, Permanent Snow and Ice, Water Bodies, Urban and Built-up Lands, Dense Forests, Open Forests, Forest - Cropland Mosaics, Natural Herbaceous, Natural Herbaceous - Croplands Mosaics, Herbaceous Croplands, Shrublands, and Unclassified)	Terra and Aqua combined Moderate Resolution Imaging Spectroradiometer (MODIS) Land Cover Type (MCD12Q1) Version 6.1
Elevation	NASADEM (through USGS OPeNDAP)

We employ an automated pipeline to fetch and update these data, both for the full history of the model training period and for monthly updates when new disease case predictions are made.

Data are fetched programmatically in Python utilizing the Climate Data Store (CDS) API, NASA Earthdata, and USGS OPeNDAP. API keys are maintained for access to the CDS API and NASA Earthdata. After download, we utilize Python libraries for data processing (including numpy, pandas, rasterio, and rasterstats) to compile and summarize data for each country, administrative zone, and timepoint.

Monthly updates of each dataset (except elevation, which is considered static and is generated once) add new data to the historical data tables that store summary statistics for each month and administrative unit. Monthly statistics represent data from the given calendar month, except for climate data in the current month which leverages the most recent 30 days of data to provide the most up-to-date data possible. Processing for each dataset is described in more detail below.

- **Climatological Variables:**

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Climate data are based on the ERA5-Land hourly dataset. This is a reanalysis dataset combining observed and modeled data into a temporally and spatially consistent global product. The data are available hourly at a spatial resolution of 0.1 degrees (approximately 9 km) and are published with approximately a 5-day lag time. From the suite of variables available, we obtain the following: 2m temperature, 2m dewpoint temperature, 10m u component of wind, 10m v component of wind, surface pressure, and total precipitation.

Data are fetched using the Python CDS API (documentation can be found here and here). To obtain the history of data for the full range of years and months for model training, we make successive calls to the CDS API using each year, month, and bounding extent of each country. Hourly data covering the month are returned for the requested variables, from which we summarize to monthly values per pixel: sum of total precipitation; mean, minimum, and maximum of 2m temperature; and mean of all other variables. Then, mean zonal statistics for each variable are calculated for each administrative zone and month using the rasterio and rasterstats Python libraries. Due to the spatial extent of ERA5-Land data cutting off near coastlines, some small administrative zones do not overlap with ERA5-Land data. In these cases, to capture approximate data using the nearest available values, a 10 km buffer is calculated around those administrative zones and this buffered area is then used for zonal statistics. Additionally, humidity is calculated for all administrative zones using the 2m temperature and 2m dewpoint temperature.

As described above, when approximating values for the current month, we utilize climate data from the most recent 30 days available. For instance, if we obtain the latest climate data on September 15, we will approximate September climate values based on data that will cover roughly August 12 to September 10 (accounting for the 5-day lag in availability). We accomplish this by requesting from the CDS API any days available in the current month and all days available in the previous month, then concatenating and slicing data arrays to extract the latest 30 days. Of note, queries to the CDS API make use of the "nocache" parameter to always obtain the latest available data for a given month. Then, going back to the example above, when running the climate data updates again during the month of October, we re-acquire September data to use the full

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calendar month of data going forward, and use the most recent 30 days to approximate October values.

- **NDVI**

Normalized Difference Vegetation Index (NDVI) data are based on Terra Moderate Resolution Imaging Spectroradiometer (MODIS) Vegetation Indices Monthly (MOD13A3) Version 6.1. This index, at 1 km resolution, provides information on monthly changes in vegetative land cover, available with a 1-2 month lag time. Data are fetched using Python scripts from the Earthdata portal for the appropriate year, month, and tile coverage for each respective timepoint and country area. Then, for each month and administrative zone, individual NDVI tiles are mosaiced and the mean zonal statistic is then calculated using the rasterio and rasterstats Python libraries.

- **Land Use/ Land Cover**

Land Use and Land Cover (LULC) data are based on Terra and Aqua combined Moderate Resolution Imaging Spectroradiometer (MODIS) Land Cover Type (MCD12Q1) Version 6.1. These data provide annual LULC classifications at 500 m resolution. Updates covering a given year are available approximately 8 months into the following year. We utilize classifications from the LCCS2 Land Use layer, labeled LC_Prop2 in the dataset. The classifications used are: Barren, Permanent Snow and Ice, Water Bodies, Urban and Built-up Lands, Dense Forests, Open Forests, Forest - Cropland Mosaics, Natural Herbaceous, Natural Herbaceous - Croplands Mosaics, Herbaceous Croplands, Shrublands, and Unclassified.

Data are fetched using Python scripts from the Earthdata portal for the appropriate year and tile coverage for each respective timepoint and country area. Then, for each year and administrative zone, individual MODIS LULC tiles are mosaiced and zonal statistics are calculated using the rasterio and rasterstats Python libraries. The zonal statistics represent the percentage of pixels in each administrative zone that belong to the above mutually exclusive categories.

- **Elevation**

Elevation data are based on NASADEM data acquired through USGS OPeNDAP. NASADEM data, published in 2021, utilizes optimized processing techniques that

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minimize data voids and extend coverage compared to previous NASA SRTM digital elevation data. Source NASADEM data are provided at 1 arc second resolution (approximately 30 m).

Elevation data are processed once for each country and administrative level, as we consider elevations time invariant at this resolution. The country extent is read from administrative boundary shapefiles, from which we use Python scripts to fetch applicable DEM elevation tiles from the USGS OPeNDAP portal. After download, individual DEM tiles are mosaiced and zonal statistics are calculated using the rasterio and rasterstats Python libraries. Zonal statistics include the mean, minimum, maximum, and standard deviation of elevation values in each administrative zone.

2.4. *Data pre-processing*

The data preprocessing procedure prepares raw dengue, demographic, and environmental datasets for the machine learning process. Although the specific models may have minor differences in requirements for preprocessing, the workflow is largely uniform and follows a structured process.

Before running the models and after initial setup, the data undergoes rigorous cleaning and integrity checks. Key steps include verifying that all regions have continuous monthly data, imputing or flagging missing dengue cases, excluding regions with only zero cases if specified, and converting date fields to proper datetime formats. Missing values are carefully handled: population and geographic gaps are filled with the last known values, social variables are assumed complete from census data, and climate variables with missing entries are imputed. At this stage, the target variable—disease rate, defined as cases per population—is constructed, and the forecast horizon is determined so that the model can correctly align input and output periods.

After the dataset is cleaned, scaling and dimensionality reduction are performed to improve computational efficiency and model performance. Disease rates are normalized using MinMax scaling, and input variables are grouped into climate, geographic, and social categories. Dimensionality reduction techniques such as PCA or Partial Least Squares (PLS) are then applied to reduce redundancy and multicollinearity while preserving essential information. Excluded variables, such as population, are scaled

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separately and stored in an independent data frame for later use, ensuring that critical features remain intact.

Additional transformations prepare the dataset for modeling. Auxiliary variables, such as month (scaled cyclically) and cases (used indirectly through case rates), are standardized. Random effects features are created to capture temporal patterns at the monthly and yearly levels. Finally, the reduced feature sets, auxiliary data, and excluded variables are merged into a single dataset, sorted by region and date, and matched with region codes. This comprehensive process results in a finalized, consistent, and well-structured dataset ready for forecasting dengue cases across different regions and time horizons.

2.5. *Training, predictions and validation*

The ensemble machine learning model is run on the first day of each month *m* to predict the number of cases in months *m+1* and *m+2*. The operational system automatically retrieves the latest trained models and runs the prediction pipeline without manual parameter adjustment. The predictors include the environmental and climatological data together with the demographic and physical data.

- ***Model retraining steps and validation***[1]

After preprocessing and finalizing the dataset, the data is divided into training and validation sets. A cutoff date is established by subtracting a configurable number of validation years (2 years) from the last available observation. Data before this cutoff is used for training, while data after the cutoff is reserved for validation. This ensures that validation is performed on more recent, unseen data, reflecting a realistic forecasting scenario. The modeling pipeline specifies that the last two years of data is reserved for model validation, and consequently the training data starts around 2010 and ends two years prior to the model training date. Retraining is useful in ensuring we avoid model drift, or the tapering of performance with an older, stale model. Additionally, regular retraining means that more recent data is shifted from the validation to the training set,

[1] Note: Models are retrained periodically — ideally every six months, or whenever new disease case data (e.g., from countries like Peru) become available. Model retraining and validation are done at the backend and not by DIRE users.

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allowing the model to learn from recent data, which includes worsening spikes in case loads.

The datasets are reshaped into arrays suitable for model input. For each region, the process loops through all months of data, using a rolling time window of past observations to predict outcomes a specified number of months ahead. This shifting-window approach generates aligned input–output pairs for both training and validation.

To improve robustness, the training and validation data are augmented. The augmentation procedure creates synthetic samples by adding small amounts of noise to the input features, effectively multiplying the dataset and reducing overfitting. Targets remain unchanged. The augmented data is then reshaped once more for compatibility: 2D arrays for CatBoost and SVM, and 3D arrays for LSTM, reflecting the specific input requirements of each model type.

During validation, the models (CatBoost, SVM, LSTM, and the Random Forest ensemble) are trained on the training set and then evaluated on the validation set. Predictions are compared with observed values to assess performance. Metrics such as Root Mean Squared Error (RMSE), Normalized RMSE (NRMSE), and R^2 are computed. Results are grouped into terciles of low, medium, and high confidence to provide interpretability. The NRMSE values (one value per region within that admin) are pulled from the metrics file (model`s output). These NRMSE values are then split into three equally sized groups (terciles). The lowest NRMSE group is assigned highest confidence as NRMSE values near 0 are optimal. Validation plots can be generated during the model training by region, showing observed versus predicted values, with vertical lines marking the training–validation cutoff (this option is not open to DIRE users).

2.6. *Resource requirements*

In addition to the predictions of the number of Dengue and Malaria cases each month, DIRE also provides estimated number (quantity) and cost of commodities and personnel that may be required in support of the impacted area. Commodities include: Vaccines doses, vaccination centers, doses of Larvicide, Liters of Adulticide, Fumigation

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kits, mosquito release containers, repellents, mosquito nets, mosquito door/window screens, awareness campaigns, diagnostic kits, general hospital rooms/boxes and ICE Rooms/boxes. For each of these commodities, DIRE provides the total number of units required, total cost, quantity per person and cost per person. For personnel, estimations are provided for Fumigation teams, education/capacitation hours, community health workers and Dengue program operation/management team

3. The DIRE UI / Dashboard

DIRE is an online, interactive dashboard that provides monthly predictions of dengue fever and Malaria cases along with estimated resource requirements. The platform currently covers Peru and Brazil and allows users to explore data at multiple administrative levels.

At its core, the dashboard features a dynamic map that visualizes predicted case loads (number of cases per administrative unit). Administrative areas with low-confidence predictions (the lowest performance as determined by the Normalized Root Mean Square Error) are marked with a hashed pattern. For Peru, the map also displays the locations of health centers, with detailed information available for each facility, including its classification, category, geographic coordinates, and official code.

Beyond the map, the dashboard presents country-level insights, such as the total number of predicted cases for the next three months, the number of confirmed cases in the most recent month with available data, and the cumulative number of cases reported in the last 12 months. Resource requirements are also summarized at the national scale, including the expected incidence rate (cases per 100,000 people), the total cost of commodities, required personnel counts, and associated personnel costs.

More granular information is available at the administrative-unit level. For each unit, the dashboard provides predictions of dengue cases for the next three months, recent confirmed case counts, and monthly breakdowns by age group (youth ages 0–19, adults, and all cases combined). Resource requirements are also forecasted at this level, offering localized estimates of personnel and commodity needs. In addition, the dashboard displays the values of the model's predictors—such as socio-economic and

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environmental indicators—that drive the case forecasts, providing transparency into the underlying modeling process. Finally, DIRE allows users to generate static PDF reports, which include both the maps and their associated statistics, making it easier to share and integrate insights into decision-making processes.

4. Implementation to Malaria

To illustrate the robustness of this model and the potential use of the prediction within the Dashboard, we use the model to predict monthly cases of Malaria. Mosly found in tropical countries and similarly to Dengue, Malaria is spread to humans by some types of mosquitoes. According to the WHO, globally in 2023, there were an estimated 263 million malaria cases and 597,000 malaria deaths in 83 countries [60]. In 2019, about 94% of the reported Malaria cases globally were recorded in the African region [61]. Malaria is also evident in countries like Peru and Brazil. Brazil has the second-highest burden of malaria in the Region of the Americas and is endemic in 8 states in the country [62]. In 2024, 138,327 malaria cases were reported in the country [63]. Within the country, Deforestation in the Amazon is thought to increase malaria. According to [64], a 1% increase in deforestation is associated with a 6.31% rise in malaria cases. In Brazil, the incidence of Malaria is almost exclusively restricted to the region of the Amazon Basin, where a number of combined factors favors disease transmission and impair the use of standard control procedures [65]. Nevertheless, Brazil has witnessed a decline in malaria cases in the Amazon region, from over 260,000 cases in 2011 to less than 130,000 cases in 2022 [66]. Overall, studies suggest significant progress in malaria control in Brazil, marked by substantial reductions in mortality over time [67].

Peru has around 17,000 cases of malaria per year, closing 2023 with more than 22,000 reported cases [68]. Of a total population of 32.2 million, 12.6 million people are at risk of malaria in Peru. There were 27,785 reported malaria cases (falciparum, vivax, and mixed) in 2022 [69]. Loreto, a region located in the Amazon basin, has reported more than 80% of malaria cases in the country [70]. Rain and flooding create numerous slow-flowing streams and swamps that encourage mosquitoes to breed. As a result, Loreto's malaria hotspots account for 96% of Peru’s malaria burden [71]. In 2019, the malaria incidence rate was 261.9 cases per 100 000 inhabitants and the mortality rate

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was 0.057 deaths per 100,000 inhabitants representing a reduction of 90.55% and 97.98%, respectively, in comparison with 1990 [72].

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